ECO526: AI for Financial Technologies

Individual Case Study Presentation Guidelines

1. Objective

Analyze how Artificial Intelligence is applied to a real-world problem in financial services. Students should demonstrate their understanding of AI technologies, evaluate business impact, identify risks, and develop evidence-based recommendations.

2. Topic Selection

Select one case study related to AI in financial technologies. Focus on a specific financial institution, fintech company, AI application, or financial industry problem. Possible topics include:

- AI for fraud detection and prevention
- AI in credit scoring and lending decisions
- Generative AI and LLMs in banking
- AI in payment processing and transaction analytics
- AI for anti-money laundering (AML) and regulatory compliance
- AI in investment management and algorithmic trading
- AI-powered customer service and financial chatbots
- AI in insurance and claims processing
- AI for personalized banking and customer segmentation
- AI in cryptocurrency and blockchain analytics
- AI governance, explainability, and ethical challenges in finance
- An alternative topic, subject to instructor approval

3. Required Presentation Structure

Part 1. Introduction and business problem

- Introduce the company or financial services context.
- Define the specific business problem and explain why it matters to financial institutions and customers.

Part 2. AI technology

- Explain the AI approach used or proposed and how the technology works.
- Discuss relevant methods such as machine learning, deep learning, NLP, or generative AI.

Part 3. Data and implementation

- Identify the required data and explain the model development or implementation process.
- Discuss evaluation metrics, deployment, and monitoring.

Part 4. Business impact

- Evaluate financial and operational benefits.
- Analyze relevant metrics such as fraud losses, approval rates, default rates, operating costs, processing time, or customer satisfaction.
- Support conclusions with real-world evidence or transparent calculations.

Part 5. Risks and limitations

- Discuss privacy, cybersecurity, bias, explainability, regulatory compliance, and other relevant risks.
- Explain how these risks could be managed.

Part 6. Practical demonstration or analysis

- Present a model, prototype, workflow, dataset analysis, worked example, or detailed technical case.
- Explain the results and limitations. Coding is encouraged where relevant but is not mandatory.

Part 7. Recommendations and conclusion

- Summarize the main findings and provide actionable recommendations.
- Explain whether the AI solution should be adopted, improved, expanded, or reconsidered.

4. Presentation Requirements

Format	Individual presentation
Duration	60 minutes, including discussion and Q&A
Slides	Approximately 20–30 content slides, excluding references
Sources	At least 8 credible sources, including academic research, company publications, industry reports, or regulatory documents
Quantitative analysis	At least one meaningful chart, metric-based comparison, calculation, or data analysis
Practical component	At least one demonstration, technical example, workflow, or worked business case
References	Cite all sources used in the presentation

Students must distinguish verified facts from assumptions and explain the limitations of their analysis.

5. Assessment Criteria

Criterion	Weight
Understanding of the business problem	15%
Understanding of AI technology	20%
Research quality and quantitative analysis	20%
Business impact and feasibility	15%
Risks, ethics, and regulatory considerations	10%
Recommendations and critical thinking	10%
Presentation quality and Q&A	10%
Total	**100%**

6. Academic Integrity

Students must cite their sources and ensure that factual claims are accurate. The use of generative AI must be disclosed where required by the instructor. AI-generated material must be independently verified, and students must be able to explain their analysis and defend their conclusions.

7. Key Expectation

The presentation must go beyond describing an AI application. Students are expected to explain how the technology works, evaluate whether it creates measurable value, identify its limitations, and develop their own evidence-based conclusions.